

1 Setting up Linux

- Windows

If you are currently on Windows 10 or newer, you are recommended to use WSL (Windows Subsystem for Linux). You may follow <https://learn.microsoft.com/en-us/windows/wsl/install> to set it up on your computer.

- Mac

If you are using MacOS, you technically don't need to install Linux since it is already Unix-based and works similarly to Linux. However, do note that there are some subtle differences (different file system layouts, BSD vs GNU etc.).

If you do want to set up Linux locally, you are recommended to do it via a virtual machine. Depending on whether you have an Intel chip or an Apple silicon chip, there are different options (e.g. VirtualBox, VMWare, UTM etc.). You should explore various options and understand their differences before committing to any one of them.

If you really want a more "permanent" version, you can consider dual-booting, which typically involves partitioning your disk drive and installing from a thumb drive. There are a lot of tutorials online, you may follow them (but make sure you know what each step is doing!)

- Linux



2 Setting up Python

In this course we will make use of various Python modules/libraries. You are advised to set up a virtual environment for easier management. We recommend using `uv` as your Python package manager. You may find the installation instructions and tutorial here: <https://docs.astral.sh/uv/getting-started/>.

Here is a non-exhaustive list of libraries (outside of the standard library) that we recommend you to install (may not be just for this course, but for scientific computing in general):

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|------------------------|---------------------------|-----------------------------|
| • <code>numpy</code> | • <code>mpi4py</code> ** | • <code>polars</code> |
| • <code>scipy</code> | • <code>matplotlib</code> | |
| • <code>numba</code> * | • <code>cython</code> | • <code>pytorch</code> * |
| • <code>cupy</code> * | • <code>pandas</code> | • <code>scikit-learn</code> |

* These packages require having an NVIDIA GPU and CUDA installed on your computer. If you don't have them, you may simply ignore them. (You can install `numba` and `pytorch` without a GPU.)

** To use `mpi4py`, you need to have an implementation of MPI installed in your system. We recommend OpenMPI.

3 Text Editor

You will also need a text editor to write (Python) programs for this course. We recommend using Visual Studio Code, but you are free to use any (vim, emacs, xcode, or even full-fledged IDEs).